and, for a vector function $\boldsymbol{w} : \mathcal{D} \to \mathbb{R}^{\mathrm{N_c}}$, we have

$$\|\boldsymbol{w}\|^2_{L^2(\mu)} = \sum_{i=1}^{\mathrm{N_c}} \|w_i\|^2_{L^2(\mu)}.$$

With this in mind, we state a useful property that we will need in the next chapter in order to clarify results on the error in invariant measures.

Proposition 6.4 *Let U be C^∞ on the position domain $\mathcal{Q} = \mathcal{T}^{\mathrm{N_c}}$ and suppose $g \in \mathcal{H}^1(\mu)$ satisfies*

$$\int_{\mathcal{D}} g \mathrm{d}\mu_\beta = 0,$$

where μ_β is the canonical measure, then the Poisson equation

$$\mathcal{L}^\dagger_{\mathrm{LD}} \varphi = g$$

has a solution $\varphi \in \mathcal{H}^1(\mu)$ which is unique up to addition of a constant multiple of the Gibbs density.

The condition is a consequence of the Fredholm Alternative which says that, for a linear operator $\mathcal{A}$ with compact resolvent, we have one of the following two conditions:

- $\mathcal{A}\varphi = g$ has a solution.
- There is a function y with $\mathcal{A}^\dagger y = 0$ and $\langle y, g \rangle \neq 0$.

In our case, $\mathcal{A} = \mathcal{L}^\dagger_{\mathrm{LD}}$. The only solutions of $\mathcal{L}_{\mathrm{LD}} y = 0$ (conserved quantities of Langevin dynamics) are the constants. So the condition of the proposition eliminates the second alternative, leaving only the first.

If we have two solutions φ_1, φ_2 of $\mathcal{L}^\dagger_{\mathrm{LD}} \varphi = g$, then their difference w must be a solution of $\mathcal{L}^\dagger_{\mathrm{LD}} w = 0$ and the only such solution is a constant multiple of the Gibbs density ρ_β.

More properties of the operator $\mathcal{L}_{\mathrm{LD}}$ are collected in [223].

6.4.5 Convergence of Averages

Given a collection of phase space points distributed according to the density $\rho_0(\boldsymbol{q}, \boldsymbol{p})$, we know that its evolution (under Langevin dynamics) after a time t will be described by

$$\rho(\boldsymbol{q}, \boldsymbol{p}, t) = e^{t\mathcal{L}^\dagger_{\mathrm{LD}}} \rho_0(\boldsymbol{q}, \boldsymbol{p}).$$